MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026
Fifth SEMESTER
Discipline Specific Elective (DSE) - MG5DSEFSC301 - DNA ANALYSIS
(2024 ADMISSION ONWARDS)

SCHEME OF VALUATION / EVALUATION

Duration: 2 hours
Maximum Marks: 70

General Instructions for Examiners:

1. Award marks for equivalent scientifically correct answers and accurate biological terminology.
2. The value points below are indicative, structured according to the syllabus and Bloom's Taxonomy levels indicated in the question paper.
3. For questions with split marks, award marks strictly in accordance with the allocated value points.
4. Under no circumstances should marks exceed the maximum mark allotted for each question or section.
5. Students are instructed to attempt at least one question from each Course Outcome (CO). Evaluate choices according to university examination rules.

Part A – One Word Answer Questions

Answer all questions. Each question carries 1 mark. Maximum: 10 Marks.

Question 1. What is the role of RNA polymerase? (1 Mark) [CO1, U]

- Synthesizes RNA from a DNA template during transcription (or transcription / RNA synthesis) **— 1 mark**

Question 2. Which biological fluid is most commonly encountered as a source of nuclear DNA in forensic investigations? (1 Mark) [CO2, A]

- Blood (liquid blood, bloodstains, dried blood) / White blood cells (leukocytes) **— 1 mark**

Question 3. Which property of DNA allows individuals to be differentiated genetically? (1 Mark) [CO2, E]

- Genetic polymorphism / sequence variation (hypervariability / VNTRs / STRs) **— 1 mark**

Question 4. What is an ethical concern? (1 Mark) [CO3, U]

- An issue or dilemma involving moral principles, rights, privacy, or societal values (e.g., unauthorized genetic surveillance, misuse of DNA databases, lack of informed consent) **— 1 mark**

Question 5. State one forensic importance of population genetics. (1 Mark) [CO1, U]

- Calculating allele frequencies / estimating match probability (random match probability) / statistical weight of DNA evidence in court **— 1 mark**

Question 6. Which index is obtained by multiplying the paternity indices obtained from individual genetic loci? (1 Mark) [CO2, E]

- Combined Paternity Index (CPI) **— 1 mark**

Question 7. Which condition should be avoided during storage of biological samples to prevent DNA degradation? (1 Mark) [CO2, E]

- Moisture / Humidity / Warm temperatures / Heat (accept: wet/damp conditions, exposure to direct sunlight/UV radiation) **— 1 mark**

Question 8. Which PCR stage separates the two strands of template DNA? (1 Mark) [CO2, A]

- Denaturation (heat denaturation at 94°C–95°C) **— 1 mark**

Question 9. Which value generally indicates stronger support for biological paternity: a higher or lower paternity index? (1 Mark) [CO2, A]

- Higher paternity index (higher value) **— 1 mark**

Question 10. What is a genetic marker? (1 Mark) [CO4, U]

- A known, identifiable DNA sequence or gene with a specific physical location on a chromosome used to track traits, inheritance, or individual variation (e.g., STR, SNP, RFLP) **— 1 mark**

Part B – Short Answer Questions

Answer any 5 questions. Each question carries 3 marks. Maximum: 15 Marks.

Question 11. Apply DNA profiling to establish the identity of a person using recovered skeletal remains in a missing-person case. (3 Marks) [CO2, A]

- Sampling and Extraction: Dense cortical bone (femur/tibia) or undamaged teeth (pulp) sampled; cleaned, ground, demineralized with EDTA, followed by organic/silica-based extraction **— 1 mark**
- Amplification and Profiling: STR profiling (mini-STRs or mitochondrial DNA sequencing / hypervariable regions HV1/HV2 if nuclear DNA is severely degraded) **— 1 mark**
- Comparison and Identity Establishment: Comparison against antemortem reference samples (personal items like toothbrush, hairbrush) or pedigree reference samples from close biological relatives (parents, children, siblings) **— 1 mark**

Question 12. Explain the inheritance pattern of Y-STR markers. (3 Marks) [CO2, An]

- Paternal Inheritance: Transmitted exclusively from father to son along the direct patrilineage without recombination across the non-recombining region of the Y-chromosome (NRY) **— 1 mark**
- Haplotype Nature: Loci are linked and inherited together as a single block (haplotype); all paternally related males share identical Y-STR profiles (barring rare mutations) **— 1 mark**

- Forensic Application and Limitation: High value in male-female mixtures (e.g., sexual assault cases) and male lineage identification; limitation is inability to differentiate between biological brothers or father and son **— 1 mark**

Question 13. Explain the role of positive and negative controls. (3 Marks) [CO2, K]

- Definition and Need: Quality assurance/quality control (QA/QC) tools essential to validate forensic testing protocols and ensure court-admissibility of DNA results **— 1 mark**
- Positive Control: A known standard DNA sample (e.g., 9947A, 007) run to verify that extraction reagents, PCR master mix, primers, polymerase, and thermal cyclers function correctly **— 1 mark**
- Negative Control / Reagent Blank: Reaction without template DNA to detect contamination, extraneous DNA introduction, or non-specific amplification across reagents or laboratory handling **— 1 mark**

Question 14. Explain the role of expert testimony in DNA evidence. (3 Marks) [CO3, An]

- Objective Presentation: Explaining complex scientific methodology, collection protocols, analytical techniques, and quality control procedures in clear, understandable language to the court/jury **— 1 mark**
- Statistical Interpretation: Presenting match probability, likelihood ratios, and Combined Paternity Indices objectively without committing the prosecutor's fallacy or defense fallacy **— 1 mark**
- Ethical and Impartial Duty: Acting as an independent and impartial minister of justice, evaluating limitations, artifact possibilities (stutter, drop-in/drop-out), and maintaining scientific integrity **— 1 mark**

Question 15. Discuss the importance of DNA discoveries in the development of forensic science. (3 Marks) [CO1, K]

- Historical Paradigm Shift: Discovery of DNA structure (Watson and Crick) and forensic profiling by Sir Alec Jeffreys (1984) revolutionized biological evidence from exclusionary blood grouping to positive individualization **— 1 mark**
- Exoneree Exoneration and Conviction Integrity: Provided an indisputable biological gold standard for solving cold cases, identifying disaster victims, and post-conviction exonerations of wrongfully convicted individuals **— 1 mark**
- Establishment of Forensic Databases: Paved the way for standardized national DNA databases (CODIS, NDNAD), automated STR typing, and enhanced speed and reliability in criminal investigations **— 1 mark**

Question 16. Discuss the usefulness of mitochondrial DNA in degraded biological samples. (3 Marks) [CO2, U]

- High Copy Number per Cell: Hundreds to thousands of mitochondria per cell compared to only two copies of nuclear DNA; significantly higher survival rate in aged, burnt, or severely degraded remains **— 1 mark**
- Suitability for Specific Matrices: Highly effective for telogen shed hairs lacking root bulbs, desiccated bone fragments, ancient skeletal remains, and teeth **— 1 mark**

- Maternal Lineage Tracing: Inherited maternally without recombination, allowing living maternal relatives spanning multiple generations to serve as reference samples (analyzing HV1 and HV2 regions) **— 1 mark**

Question 17. Explain the importance of traceability in DNA analysis. (3 Marks) [CO2, K]

- Chain of Custody: Complete, unbroken chronological documentation tracking evidence collection, custody, transfer, analysis, storage, and court submission to prevent tampering or substitution **— 1 mark**
- Reagent and Equipment Calibration: Traceability of all chemical reagents, lot numbers, pipettes, and thermal cyclers to certified reference standards under ISO/IEC 17025 accreditation **— 1 mark**
- Reproducibility and Auditability: Enables independent re-testing, validation of standard operating procedures (SOPs), and defense scrutiny, ensuring courtroom legal defensibility **— 1 mark**

Part C – Short Essay Questions

Answer any 5 questions. Each question carries 6 marks. Maximum: 30 Marks.

Question 18. Apply the maternity index to determine the biological relationship between a mother and child in a disputed maternity case. (6 Marks) [CO2, A]

- Principle and Concept of Maternity Testing: Explanation of Mendelian inheritance; a child inherits one obligate allele from the biological mother at every autosomal STR locus. Disputed maternity scenarios (baby switching, illegal adoption, immigration verification) **— 1.5 marks**
- Maternity Index (MI) Calculation: Formulating the likelihood ratio (LR) at each locus: MI = P(Child genotype | Alleged Mother is true mother) / P(Child genotype | Random unrelated woman is mother). Formula based on allele transmission and population allele frequencies **— 1.5 marks**
- Combined Maternity Index (CMI) and Probability of Maternity (W): Calculation of CMI = Product of individual locus MIs (CMI = MI1 × MI2 × ... × MIn). Conversion into Probability of Maternity using Prior Probability: W = (CMI) / (CMI + 1) assuming prior odds of 0.5 (Essen-Möller equation) **— 1.5 marks**
- Forensic Interpretation and Exclusion Rules: Establishing exclusion (two or more locus incompatibilities not explained by mutation) vs. inclusion (CMI > 10,000, Probability > 99.99%); handling possible single-locus mutations **— 1.5 marks**

Question 19. Analyse the major limitations of DNA profiling. (6 Marks) [CO4, An]

- Sample Degradation and PCR Inhibition: Environmental exposure (heat, moisture, UV, soil microbes) causes strand shearing; common inhibitors (haem, indigo dyes, humic acid, calcium) cause PCR amplification failure or allelic drop-out **— 1.5 marks**
- Low Template DNA (LTDNA) and Artifacts: Touch DNA and sub-picogram amounts produce technical artifacts: stochastic allelic drop-out, allelic drop-in, elevated stutter peaks, and heterozygote peak height imbalance **— 1.5 marks**

- Complex Mixtures and Deconvolution Challenges: Presence of three or more biological contributors, unequal contributor ratios, masking of minor donor alleles under major donor stutter peaks leading to ambiguous profiles **— 1.5 marks**
- Legal, Biological, and Identical Twin Limitations: Inability of standard STR profiling to distinguish monozygotic (identical) twins; non-recombining lineage markers (Y-STR/mtDNA) cannot isolate specific individuals within a lineage; ethical/privacy and genetic database surveillance concerns **— 1.5 marks**

Question 20. Explain the determination of allele frequencies using population data. (6 Marks) [CO2, An]

- Concept of Population Genetics in Forensics: Necessity of population-specific reference databases; an allele match is statistically meaningful only when evaluated against the background prevalence of that allele in the relevant ethnic group **— 1.5 marks**
- Sampling Methodology and Representative Cohort: Collecting representative, unrelated individuals from clearly defined geographic/ethnic populations; adhering to sample size adequacy (minimum 100–200 individuals / 200–400 chromosomes) to reliably capture rare alleles **— 1.5 marks**
- Mathematical Determination (Gene Counting Method): Direct counting formula: $p_i = (2 * N_{ii} + N_{ij}) / (2 * N)$, where N_{ii} is the count of homozygotes for allele i, N_{ij} is heterozygotes, and N is the total sample size **— 1.5 marks**
- Hardy-Weinberg Equilibrium (HWE) and Independence: Testing population data for HWE ($p^2 + 2pq + q^2 = 1$) using chi-square or exact tests; verifying Linkage Equilibrium (LE) between distinct STR loci to justify applying the product rule for cumulative match probability **— 1.5 marks**

Question 21. Discuss the importance of human genetics in forensic science and biological identification. (6 Marks) [CO1, U]

- Biological Basis of Individuality: The human genome consists of 3.2 billion base pairs; 99.9% identical, but the 0.1% polymorphic variation accounts for unique genetic identity (except monozygotic twins) **— 1.5 marks**
- Mendelian Inheritance and Lineage Tracing: Application of autosomal segregation laws to solve disputed parentage (paternity/maternity) and missing persons; non-Mendelian maternal inheritance of mtDNA and paternal patrilineal inheritance of Y-chromosome markers **— 1.5 marks**
- Forensic DNA Phenotyping and Biogeographical Ancestry: Utilization of human genetic markers (e.g., SNPs, Indels) to infer physical phenotypic traits (pigmentation of eyes, hair, skin) and ancestral origin in investigative leads for suspect-less cases **— 1.5 marks**
- Legal, Disaster Victim Identification (DVI) and Mass Casualty Impact: Role of population genetics and forensic statistics in disaster victim identification, mass graves, human trafficking, and establishing evidence admissibility in criminal jurisprudence **— 1.5 marks**

Question 22. Describe the role of STRs in DNA profiling and their importance in forensic identification. (6 Marks) [CO2, K]

- Definition and Structure of STRs: Microsatellites / Short Tandem Repeats consisting of 2 to 6 base pair core repeat units (e.g., tetranucleotides like GATA) tandemly repeated in non-coding regions across the genome **— 1.5 marks**
- Forensic Advantages of STRs: Small amplicon size (100–450 bp) ideal for degraded samples; discrete allele designations based on integer repeat counts; low mutation rates suitable for forensic tracking **— 1.5 marks**
- Multiplex PCR and Capillary Electrophoresis: Simultaneous fluorescent amplification of 15–24 loci in a single tube; high-resolution separation and detection via multi-capillary electrophoresis with allelic ladders and internal size standards **— 1.5 marks**
- Forensic Significance, CODIS and Discriminatory Power: Core loci utilized in national databases (CODIS 20 core loci); extremely high discriminatory power where cumulative random match probability exceeds 1 in a billion or trillion, providing definitive individualization **— 1.5 marks**

Question 23. Define DNA and explain its chemical composition and double-helical structure. (6 Marks) [CO1, U]

- Definition and Cellular Organization: Deoxyribonucleic acid is the primary hereditary biomolecule located in the nucleus (nuclear DNA) and mitochondria (mtDNA) that encodes genetic instructions **— 1.5 marks**
- Chemical Building Blocks (Nucleotides): Composed of repeating nucleotide units, each comprising: (a) Deoxyribose sugar, (b) Phosphate group, and (c) Four nitrogenous bases: Purines (Adenine [A], Guanine [G]) and Pyrimidines (Cytosine [C], Thymine [T]) **— 1.5 marks**
- Watson-Crick Double Helical Model: Two antiparallel polynucleotide strands wound around a central axis (5' to 3' and 3' to 5'); sugar-phosphate covalent backbone joined by 3',5'-phosphodiester bonds; major and minor grooves **— 1.5 marks**
- Complementary Base Pairing and Structural Stability: Specific complementary hydrogen bonding: Adenine pairs with Thymine via two hydrogen bonds (A=T), Guanine pairs with Cytosine via three hydrogen bonds (G≡C); hydrophobic base stacking forces providing thermal and structural stability **— 1.5 marks**

Question 24. Evaluate the suitability of hair as a source of DNA evidence and discuss the factors affecting DNA recovery. (6 Marks) [CO2, E]

- Morphology and Growth Phases of Hair: Anagen (active growth, cellular follicular tag rich in nucleated cells), Catagen (transition phase), and Telogen (shed/quiescent phase, heavily keratinized with minimal/no nucleated pulp cells) **— 1.5 marks**
- Nuclear vs. Mitochondrial DNA Suitability: Hairs with roots/follicular tags yield abundant nuclear DNA suitable for autosomal STR profiling; shed telogen hairs lack nuclear DNA and must be routed for mitochondrial DNA (mtDNA) sequencing (HV1/HV2) **— 1.5 marks**
- Extraction and Lysis Protocols: Complete digestion of tough keratinized protein matrices requires specialized chemical digestion: Proteinase K and strong reducing agents (DTT - Dithiothreitol) to break disulfide bonds **— 1.5 marks**

- Factors Affecting Recovery and Forensic Limitations: Environmental weathering (UV sun exposure, microbial rot); cosmetic treatments (chemical dyes, bleaches, perming agents act as PCR inhibitors or destroy DNA); root shearing/loss during forceful extraction; lack of individualizing power if only mtDNA is recovered **— 1.5 marks**

Part D – Essay Questions

Answer any 1 question. Each question carries 15 marks. Maximum: 15 Marks.

Question 25. Define DNA variation and explain polymorphism, single nucleotide polymorphism and their importance in genetic analysis. (15 Marks) [CO1, U]

- Definition of DNA Variation and Genetic Diversity: Concept of sequence variation arising from mutations, insertions, deletions, and structural rearrangements across the human genome; distinction between coding and non-coding variations **— 3 marks**
- Concept and Types of Genetic Polymorphism: Definition of polymorphism (existence of two or more alleles in a population with frequency > 1%); Length polymorphisms (VNTRs, minisatellites, STRs/microsatellites) vs. Sequence polymorphisms; mechanisms generating variability (replication slippage, unequal crossing-over) **— 4 marks**
- Single Nucleotide Polymorphisms (SNPs) - Characteristics and Detection: Definition of SNPs (single base changes A/T/G/C); bi-allelic nature; low mutation rates ($\sim 10^{-8}$); distribution across the genome (over 10 million SNPs); modern detection technologies (microarrays, TaqMan assays, Next-Generation Sequencing / MPS) **— 4 marks**
- Forensic, Medical, and Evolutionary Importance: Forensic individualization of severely degraded samples using mini-SNPs; Biogeographical Ancestry (AIMs - Ancestry Informative Markers); Phenotypic SNP profiling (HIrisPlex for eye/hair color); pharmacogenomics and evolutionary phylogenetics **— 4 marks**

Question 26. Evaluate the historical development of DNA profiling and critically discuss its importance and applications in modern forensic science. (15 Marks) [CO2, E]

- Historical Milestones and Genesis of DNA Profiling: Discovery of hypervariable minisatellites and development of multi-locus probe (MLP) DNA fingerprinting by Sir Alec Jeffreys (1984); Colin Pitchfork case (first criminal conviction and first exoneration via DNA, 1986); transition to Single Locus Probes (SLP) and RFLP-Southern Blotting (principles, limitations of high template requirement and radioactive detection) **— 3.5 marks**
- Technological Revolution - PCR, STRs, and Capillary Electrophoresis: Introduction of Polymerase Chain Reaction (PCR) by Kary Mullis; evolution through HLA-DQA1 / Polymarker dot-blots; development of automated fluorescent Short Tandem Repeat (STR) typing; multiplex PCR systems and multi-capillary electrophoresis with laser-induced fluorescence **— 4 marks**
- Modern Applications in the Criminal Justice System: Violent crime investigation (homicide, sexual assault); Mass Disaster Victim Identification (DVI - tsunami, 9/11,

aviation crashes); disputed paternity and kinship analysis; human trafficking, missing persons tracing, and Cold Case resolution **— 4 marks**

- Critical Evaluation, Advanced Frontiers, and Contemporary Challenges: Establishment of national DNA databases (CODIS, NDNAD); emergence of Next-Generation Sequencing (NGS/MPS), Rapid DNA systems, and Investigative Genetic Genealogy (IGG); critical examination of challenges: low-template DNA artifacts, complex DNA mixtures, contamination risks, secondary transfer, laboratory error rates, ethical/privacy concerns, and genetic surveillance **— 3.5 marks**